QUERY-INDEPENDENT LEARNING FOR VIDEO SEARCH

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ABSTRACT

Most of existing learning-based methods for query-by-example take the query examples as "positive" and build a model for each query. These methods, referred to as query-dependent, only achieved limited success as they can hardly be applied to real-world applications, in which an arbitrary query is usually given. To address this problem, we propose to learn a query-independent model by exploiting the *relevance* information which exists in the pair of query-document. The proposed approach takes a query-document pair as a sample and extracts a set of query-independent textual and visual features from each pair. It is general and suitable for a real-world video search system since the learned relevance relation is independent on any query. We conducted extensive experiments over TRECVID 2005-2007 corpus and shown superior performance (+37% in Mean Average Precision) to the query-dependent learning approaches.

1. INTRODUCTION

Due to the explosive growth of video data (both online and offline), there has been increasing expectation for video search technologies to effectively and efficiently index and retrieve the videos. A typical video search system is usually designed on the basis of the similarity between the query and documents (e.g., shots) in terms of text or visual content. For example, Mei *et al.* [1] extracted a set of low-level features from each query example and shot in the entire video corpus, and return the most similar shots to the query. However, one problem with typical search systems is the heavy reliance on the exact match or feature similarity for judging semantic relevance.

An alternative to video search [2][3][4] applied supervised learning algorithms to learn a specific model for each query by taking the query examples as "positive". The search is posed as a classification problem, aiming at identifying a video shot relevant (i.e., "positive") or irrelevant (i.e., "negative") to a set of predefined queries. For example, Natsev *et al.* [2] used low-level visual features to model the semantics of queries by smart data sampling and selection. Chang *et al.* [3] extracted textual features from Automatic Speech Recognition (ASR) results such as *tf-idf*, to train query models for improving the performance of text baseline search. Tešić *et al.* [4] also used the shot scores predicted by pre-trained concept models as the semantic features to learn the semantics of queries. However, in real-world video search applications, the queries submitted by users are usually arbitrary rather than predefined. The learned classifier for one query cannot be applied to the others. Therefore, the query-dependent learning approaches achieved only limited success for video search. The upside of Figure 1 illustrates the diagram of the query-dependent learning.

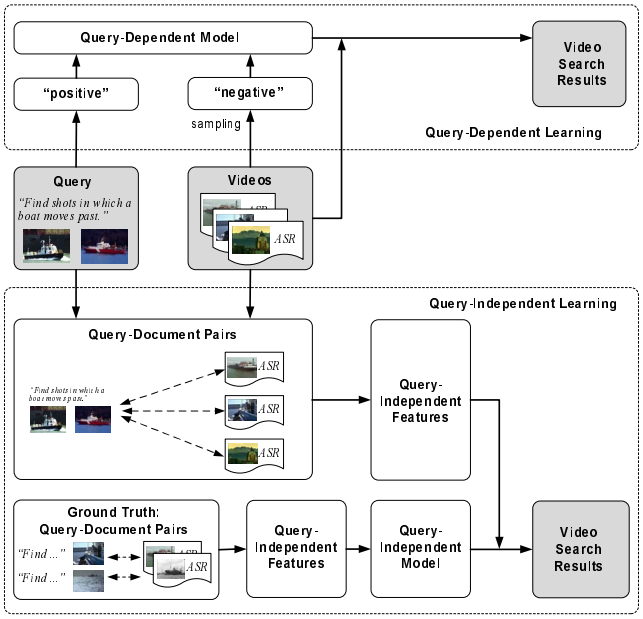

Fig. 1: The diagrams of query-dependent and query-independent learning for video search.

To address the query-dependent problem, we propose to build *query-independent* model for video search. This model learns the relevance relation in *query-document* pairs collected from the search history into account, thus it is built independently on the new query submitted to the search system. Specifically, the proposed approach takes a query-document pair as a sample and extracts a set of query-independent tex-

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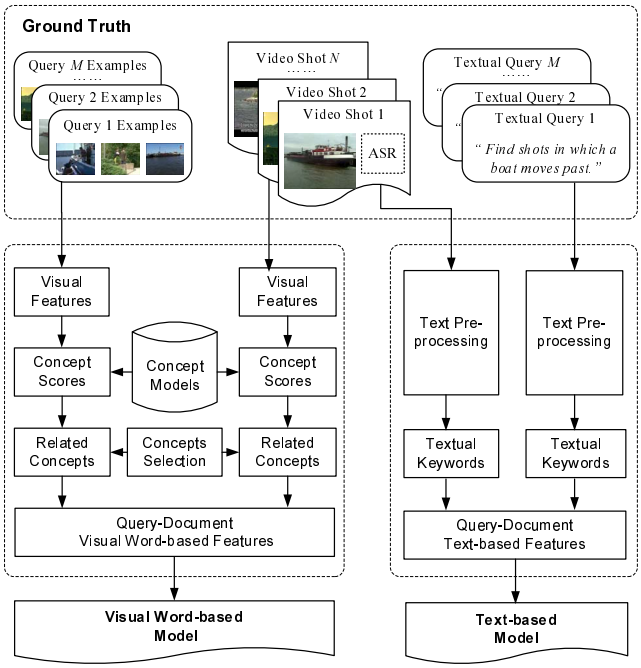

Fig. 2: The training procedure of query-independent learning.

tual and visual features from each pair. A general relevance model is then built by Support Vector Machine (SVM). As shown in the downside of Figure 1, this query-independent approach is different from the traditional query-dependent approaches in terms of the training samples and features used for building the model. We argue that the features should indicate the *relevance* relation between a pair of query (textual keywords and visual examples) and document (shots in video search task). In the other word, the features measuring such relevance relation are extracted from each query-document pair, rather than from each query or document.

The rest of this paper is organized as follows. Section 2 presents the query-independent learning approach. Section 3 shows the experiments over TRECVID benchmark, followed by conclusions in Section 4.

2. QUERY-INDEPENDENT LEARNING FOR VIDEO SEARCH

As aforementioned, the most differences between *query-dependent learning* and the proposed *query-independent learning* approach, is that how to represent and learn the relevance information. Thus we have different training samples collection and different features used in learning.

2.1. Query-Independent Learning Framework

As any other supervised learning methods, our supervised video search approach includes training and test stages.

The training procedure of the proposed query-independent learning is demonstrated in Figure 2. As can be observed in this figure, the following items should be noted:

a) Two types of features are extracted, i.e., visual word-based features and text-based features.

b) All the features are extracted from query-document pairs. Visual word-based features correspond to pairs of query examples and shot keyframes, while text-based features correspond to pairs of textual queries and shot ASR.

c) There are different pre-processing before extracting the two types of features. To extract visual word-based features, we rely on pre-trained concept models to select a set of related concepts from each query example and each shot. To extract text-based features, we exploit some text processing technologies to extract textual keywords from each textual query and each ASR of shot [1].

d) The models could be built using any state-of-the-art classifiers, we use SVM in our system, due to its effectiveness in training semantic models.

In test stage, given a new query, each shot in the video corpus is combined with it to construct a query-document pair, and then a set of similar features are extracted as in the training stage. Finally, the relevancy of each pair is predicted with the search model learned in training stage.

2.2. Query-Independent Features

2.2.1. Related Concepts Selection

The visual word-based features are relied on a set of concept detectors. As the first step, we build concept models for all the annotated concepts of the LSCOM-lite [5] lexicon using SVM [1]. Second, concept scores of each shot are produced by the concept models, indicating the presence of the corresponding concepts. Then we select the concepts if their scores are higher than the corresponding thresholds, which can be determined by the concept distributions. Given a concept, we sort the scores of all the shots in video corpus in descending order and select the score ranked j as the threshold, where $j = p\,(y = +) \times M$, $p\,(y = +)$ is prior probability and M is the number of the samples in the video corpus.

2.2.2. Text-based Features and Visual Word-based Features

The text-based features indicate the relationship between textual query and shot ASR, while the visual word-based features indicate the relationship between query examples and shot keyframes. Li *et al.* [6] proposed feature *c-tf-idf* in concept space by extending the feature *tf-idf* in text domain. Here, we extend this idea and further propose more features in concept space and apply them to our learning-based video search. Related concepts are viewed as visual terms for the query examples and visual documents (shot key frames) in the corpus. For queries containing multiple examples, we combine the visual terms using OR way. For example, there are two exam-

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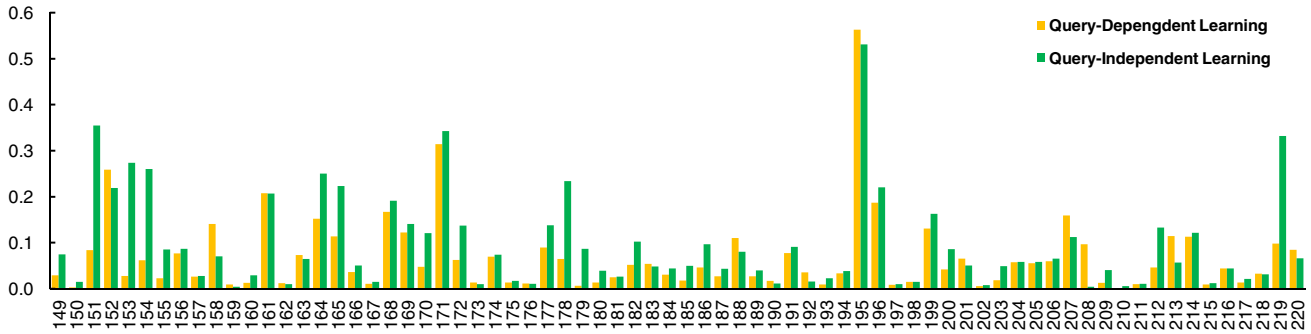

Fig. 3: AP of each query topic over TRECVID 2005-2007 test data set.

ples to a given query, and their related concepts are "*Person, Face, Animal*" and "*Animal, Road*" respectively, and then the final related concepts are "*Person, Face, Animal, Road*".

Given a document collection $D = (d_1, d_2, \cdots d_M)$ and query Q containing the term $T = (t_1, t_2, \cdots t_N)$, where M denotes the number of documents in the video corpus, N denotes the number of terms in the query.

Term frequency (*TF*): The more often query terms occur in a document; the more likely the document is to be relevant to the query. Term frequency is such a measure to relate query and document, it is given by:

$$TF(j) = \sum\nolimits_{i=1}^{N} tf\,(i,j) \tag{1}$$

where $tf(i,j)$ denotes the number of occurrences of term t_i in document d_j. When $tf(i,j)$ is used in visual space, it denotes the score of visual term in visual document. If a visual term is not the related concept of the visual document, the score is zero.

To produce other more effective features, we introduce *IDF*(Inverse document frequency). It was developed based on the intuition that a query term which occurs in many documents is not a good discriminator.

$$IDF = \sum\nolimits_{i=1}^{N} idf\,(i) = \sum\nolimits_{i=1}^{N} \log \frac{M + 0.5}{m\,(i) + 0.5} \tag{2}$$

where $m(i)$ is the number of documents which term t_i occurs in, 0.5 is a smoothing correction used to avoid denominator is equal to 0. When $m(i)$ is used in visual space, it denotes the sum score of term t_i in the entire collection. Note that *IDF* is a constant to a given query.

TFIDF, BM25: Some features are obtained by combining *TF* and *IDF*, as well as document length $dl(j)$ (the total number of terms occurrences in document d_j). For example, *TFIDF* and *BM25* are such kind of features defined as:

$$TFIDF(j) = \sum\nolimits_{i=1}^{N} tf\,(i,j) \times idf\,(i) \tag{3}$$

$$BM25(j) = \sum\nolimits_{i=1}^{N} \frac{tf\,(i,j) \times idf\,(i) \times (k+1)}{k \times ((1-b) + b \times ndl\,(j) + tf\,(i,j))} \tag{4}$$

where $ndl(j)$ is normalized document length, defined with the average length of documents $\bar{dl}$: $ndl(j) = dl(j)/\bar{dl}$. k and b are tuning constants, k modifies the extent of the influence of term frequency. Higher values would increase the influence of *TF*. The constant b, which ranges between 0 and 1.0, modifies the effect of document length. We simply set $k = 2.0$ and $b = 0.75$ according to the results of [7].

Query term density (*QD*): For a query contains multiple terms, the more different terms occur in a document, the more likely it is relevant to the query. *QD* is defined as the number of different query terms in document d_j.

$$QD(j) = \sum\nolimits_{i=1}^{N} q\,(i,j) \tag{5}$$

where $q(i,j)$ is a binary function to indicate whether term t_i occurs in document d_j.

Some other special features can be also introduced using some special technologies, such as features indicate whether contain named entity, and features from speech classification. Meanwhile, some image/video processing technologies such as face detection, scene classification, can be also applied to extracting visual features.

2.3. SVM Learning

After extracting text-based and visual word-based features, we use SVM to train query-independent models [8]. The samples in learning are a set of query-document pairs collected from the search history, and the labels are based on the relevancy of the documents to the queries.

Compared with the query-dependent learning, the proposed query-independent learning approach will not suffer from the low number of training samples. All the queries previously submitted to the search system can be used in training search model.

3. EXPERIMENTS

We conducted the experiments on automatic search task over TRECVID 2005-2007 test set [9]. There are 24 query topics provided by NIST with ground truth of relevance in each year. In total, we have 78,419, 88,108, and 87,595 query-document

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Table 1: Experimental results for fusion of two types of features over TRECVID 2005-2007 test data sets and query topics.

METHOD	QD Learning	QInD Learning	
EVALUATION	MAP	MAP	Gain
2005	0.0878	0.1356	+54.4%
2006	0.0668	0.0863	+29.2%
2007	0.0542	0.0648	+19.6%
AVERAGE	0.0696	0.0956	+37.4%

pairs with relevance label ("positive" or "negative") in the year of 2005, 2006, and 2007, respectively. The proposed approach was evaluated on the data set of one year, while the data of the other two years were in turn used for training and validation. Consequently, there were two results for each year. Then, these two results are fused in average. LIBSVM and RBF kernels [8] are used in our experiments. The models were built upon training set, the parameters were tuned in validation set, while the performance is evaluated on test set. As more than 90% samples of each query are negative, we randomly sampled the negative samples of each query in the training data set as [10]. We used only common statistics features such as *tf* and the combinations of *tf* and *idf*, which have presented above. We took $\log(x + 1.0)$ function on all the feature values in order to reduce the effects of large numbers, where x is the original feature values.

To demonstrate the effectiveness of the proposed query-independent learning, we compare it with the query-dependent learning. The two kinds of features are extracted to build the models respectively. Then the scores produced by the two kinds of models are fused in averaging way. From Table 1, we see that the proposed QInD learning method outperform QD learning method in all data sets, and the improvements are significant, with gain more than 37% in average. Fig.3 illustrates the comparison of the proposed QInD learning and the QD learning on each query topic over TRECVID 2005-2007 test data sets. We can easily find that most of query topics (more than 75%) have better performance in QInD learning method, only a few query topics which have very strong visual coherence and homogeneity outperform the QInD learning method, such as *Topic 0158* (helicopter), *Topic 0195* (soccer), *Topic 0207* (waterfront) and *Topic 0208* (street at night). Considering that most real-world queries are of great diversity, we can conclude that our proposed QInD learning method is more robust and reliable.

4. CONCLUSIONS

In this paper, we have presented a novel query-independent learning approach to video search by investigating relevance relations from query-document pairs. Unlike conventional query-dependent learning approaches, it is more general and suitable for real-world search applications. The comprehensive experiments have proved that the proposed query-independent learning approach significantly outperforms the state-of-the-art query-dependent learning approaches. Our future work will focus on seeking more effective features which better describe the query-document relationship, as well as incrementally learning the *relevance* model based on a large collection of search results.

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